Polynomial Systems and the Field of Witnesses

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Abstract

We combine s polynomial equations over a number field into one without introducing any coordinates, with degree at most $\lceil\sqrt{s}\rceil$ times the largest original degree. The construction uses a homogeneous anisotropic form obtained from multiplication determinants at one finite prime. Over a formally real field the familiar sum of squares gives degree at most twice the largest original degree, independently of s. We also prove that an existential formula with infinitely many inputs in a number field acquires all but finitely many base-field inputs after extensions of uniformly bounded degree. For a single nonzero polynomial this degree bound can be taken to be its smallest positive individual degree in a witness coordinate. Consequently one unchanged existential formula cannot define the rings of integers, or the corresponding rings of S-integers, in all finite extensions of a number field.

1 Combining equations without new coordinates

For a field K, let $F_1, \dots, F_s \in K[X_1, \dots, X_n]$, where $s \geq 1$. The common zero set is evaluated on K^n. We seek a polynomial P in the same coordinates satisfying

$$P(\mathbf{x}) = 0 \quad \Longleftrightarrow \quad F_1(\mathbf{x}) = \cdots = F_s(\mathbf{x}) = 0 \qquad (\mathbf{x} \in K^n).$$

The equivalence concerns points over K; an extension of the field changes the assertion unless its proof also applies there.

Proposition 1.1. *If K is formally real, the zero set of $\sum_{i=1}^{s} F_i^2$ in K^n is the common zero set of $F_1, \dots, F_s$. Its degree is at most $2 \max_i \deg F_i$.*

Proof. Formal reality means that -1 is not a sum of squares in K. If a vanishing sum of squares had a nonzero summand, division by that square would express -1 as a sum of the other squares, a contradiction. Squaring gives the degree bound. □

Every number field with a real embedding is formally real. Over other number fields an anisotropic form combines the equations. A homogeneous form $A \in K[Z_1, \dots, Z_r]$ is called *anisotropic over K* if its only zero in K^r is the zero vector.

Lemma 1.2. *Let K be a number field and $e \geq 1$ an integer. There is an explicit homogeneous anisotropic form*

$$A_e \in \mathcal{O}_K[Z_{j,i} : 0 \leq j, i < e]$$

of degree e in e^2 coordinates.

Proof. Choose a nonzero prime ideal $\mathfrak{p}$ of $\mathcal{O}_K$, normalize $v_\mathfrak{p}$ to have value group $\mathbb{Z}$, and choose an element $\pi \in \mathfrak{p} \setminus \mathfrak{p}^2$ and a monic polynomial

$$g(U) \in \mathcal{O}_K[U], \qquad \deg g = e,$$

whose reduction modulo $\mathfrak{p}$ is irreducible. Such a polynomial is obtained by lifting the coefficients of an irreducible degree-e polynomial over the finite residue field.

In the e-dimensional K-algebra $B = K[U]/(g)$, use the basis $1, u, \dots, u^{e-1}$, where u is the image of U. Define

$$N_g(Z_0, \dots, Z_{e-1}) = \det\left(m_{\sum_{i=0}^{e-1} Z_i u^i}\right),$$

where m_b denotes multiplication by b in this basis. Reduction by the monic relation $g(u) = 0$ gives a matrix whose entries are linear forms in the Z_i with coefficients in $\mathcal{O}_K$. Thus N_g is a homogeneous integral form of degree e, with $N_g(1, 0, \dots, 0) = 1$.

Write $v = v_{\mathfrak{p}}$. For any nonzero vector $\mathbf{z} \in K^e$, put $a = \min_i v(z_i)$. The entries of $\pi^{-a}\mathbf{z}$ belong to the valuation ring and are not all in its maximal ideal. Modulo that ideal, the multiplication matrix is multiplication by a nonzero element of the field

$$(\mathcal{O}_K/\mathfrak{p})[U]/(\overline{g}).$$

Its determinant is nonzero in the residue field. Homogeneity therefore proves the exact valuation identity

$$v(N_g(\mathbf{z})) = e \min_i v(z_i). \tag{1}$$

In particular, $N_g(\mathbf{z}) \neq 0$ for every nonzero $\mathbf{z} \in K^e$.

Now set

$$A_e((Z_{j,i})) = \sum_{j=0}^{e-1} \pi^j N_g(Z_{j,0}, \dots, Z_{j,e-1}). \tag{2}$$

At a nonzero input, omit any zero blocks. By (1), the valuations of the remaining summands are congruent to their distinct indices j modulo e. Exactly one summand has least valuation. The sum cannot vanish. This proves anisotropy. Homogeneity and the coefficient assertion follow directly from (2). □

Theorem 1.3. *Let K be a number field and let $F_1, \dots, F_s$ be polynomials in a fixed set of coordinates, each of degree at most d. Put $e = \lceil \sqrt{s} \rceil$. Padding the list of equations with zeros to length e^2, the polynomial*

$$P = A_e(F_1, \dots, F_s, 0, \dots, 0)$$

has the same K-rational zero set as the original system, uses the same coordinates, and has degree at most ed. If the F_i have coefficients in $\mathcal{O}_K$, then so does P.

Proof. The form A_e vanishes only at the zero vector over K. Substituting the F_i therefore gives the stated equivalence. Since A_e is homogeneous of degree e, each resulting term has degree at most ed. No coordinate is introduced by the substitution. □

For example, any four equations of degree at most d over a number field admit one equivalent equation of degree at most $2d$, and any nine admit one of degree at most $3d$. For a formally real number field, Proposition 1.1 gives the stronger bound $2d$ for arbitrary s. For $s = 1$ one uses the original polynomial.

A concrete quadratic form

Over $K = \mathbb{Q}(i)$, take the prime ideal (3), whose residue field has nine elements. The element $1 + i$ is not a square in that field, since $(1 + i)^4 = -1 \neq 1$ modulo 3. Applying the construction with $e = 2$, $\pi = 3$ and $g(U) = U^2 - (1 + i)$ gives

$$A_2(Z_1, Z_2, Z_3, Z_4) = Z_1^2 - (1 + i)Z_2^2 + 3Z_3^2 - 3(1 + i)Z_4^2.$$

It is anisotropic over $\mathbb{Q}(i)$ by the valuation proof above. It combines four equations into one even though a plain sum of squares is unsuitable over that field.

For a cubic form in nine coordinates over the same field, use $\mathfrak{p} = (1+i)$, $\pi = 1+i$ and $g(U) = U^3 + U + 1$. The reduction of g has no root in $\mathbb{F}_2$ and is therefore irreducible. The multiplication determinant is

$$\begin{aligned} N(a,b,c) &= \det \begin{pmatrix} a & -c & -b \\ b & a-c & -b-c \\ c & b & a-c \end{pmatrix} \\ &= a^3 - 2a^2c + ab^2 + 3abc + ac^2 - b^3 - bc^2 + c^3. \end{aligned}$$

Consequently

$$N(Z_1, Z_2, Z_3) + (1+i)N(Z_4, Z_5, Z_6) + 2iN(Z_7, Z_8, Z_9)$$

is an explicit anisotropic cubic over $\mathbb{Q}(i)$. Its coefficient list and the determinant identity are recorded in the accompanying exact computational data.

Comparison with one norm

One can also choose an extension L/K of degree s and a K-basis $b_1, \dots, b_s$. Then

$$N_{L/K}\left(\sum_{i=1}^{s} b_i F_i\right) = 0$$

is equivalent to all $F_i = 0$, because the norm of a nonzero field element is nonzero. This gives degree at most sd. Theorem 1.3 obtains the square-root bound by combining several smaller multiplication determinants at distinct valuation classes.

2 Existential formulas after finite extensions

Fix a number field k. An existential formula considered here has the form

$$\Phi(T): \quad \exists Y_1, \dots, Y_m \quad \theta(T, Y_1, \dots, Y_m),$$

where θ is a finite Boolean combination of polynomial equations with coefficients in k. Equivalently, it is a finite disjunction of conjunctions of equations and nonzero conditions. For each extension field L/k, write

$$\Phi(L) = \{a \in L : \Phi(a) \text{ holds in } L\}.$$

The same formula and the same coefficients are used for every L.

Theorem 2.1. *If $\Phi(k)$ is infinite, then there exist a positive integer D and a nonzero polynomial $h \in k[T]$ such that every $a \in k$ with $h(a) \neq 0$ belongs to $\Phi(L_a)$ for some finite extension L_a/k of degree at most D.*

Proof. The conditions defining the witness tuples give a constructible subset of affine space. By Chevalley's theorem its image in the input line is constructible [1]. Since the image has infinitely many k-points, it contains the generic point of the line. Equivalently, at least one of the finitely many conjunctions has a nonempty generic fiber over $k(T)$.

Choose a closed point in a nonempty affine open of that generic fiber. Its residue field E is a finite extension of $k(T)$. The coordinates of the point give elements $y_1, \dots, y_m \in E$ satisfying one conjunction, with every required nonzero expression nonzero in E. Characteristic zero gives a primitive element α for $E/k(T)$. Let $G(T, Z) \in k(T)[Z]$ be its monic minimal polynomial, of degree $D = [E : k(T)]$ in Z.

Express each y_i, and also the inverse of each required nonzero expression, as a polynomial in α of degree less than D with coefficients in $k(T)$. Choose a nonzero $h \in k[T]$ which clears all denominators in these expressions and in G. Put $R = k[T, 1/h]$ and

$$C = R[Z]/(G).$$

Because G is monic, C is free of rank D over R and embeds in E under $Z \mapsto \alpha$. The chosen coordinate expressions and inverse expressions belong to C. All the equations, and the identities asserting that the chosen inverses are inverses, hold in C: they hold in E, and $C \hookrightarrow E$ is injective.

Now let $a \in k$ with $h(a) \neq 0$. Specialization gives the nonzero D-dimensional algebra

$$C_a = k[Z]/(G(a, Z)).$$

Choose a maximal ideal of C_a. Its residue field L_a is a finite extension of k of degree at most D. The coordinate expressions descend to a witness tuple in L_a; all equations and inverse identities survive. Hence the same conjunction, and therefore $\Phi(a)$, holds in L_a. □

The finite exceptional set in the theorem consists of the k-roots of the single fixed polynomial h. The degree bound D is uniform in a, although the extension L_a may depend on a. The proof specifies D as the degree of one generic closed point; it does not identify one extension that contains witnesses for all inputs.

Corollary 2.2. *No single existential formula over a number field k defines $\mathcal{O}_L$ in every finite extension L/k. More precisely, if $\Phi(k)$ is infinite, there is a positive integer D such that, for all but finitely many $q \in \mathbb{Q} \setminus \mathbb{Z}$, the formula accepts q in some extension L_q/k of degree at most D.*

Proof. Apply Theorem 2.1 to the elements $q \in \mathbb{Q} \subset k$. Every rational algebraic integer is an integer. Indeed, if $q = a/b$ is in lowest terms and satisfies a monic equation of degree r with integer coefficients, clearing denominators yields $b \mid a^r$, so $b = 1$. Consequently each admitted rational noninteger lies outside $\mathcal{O}_{L_q}$. □

Corollary 2.3. *Let S be a fixed finite set of finite primes of k, and for each finite extension L/k let S_L be the primes of L above S. One unchanged existential formula over k cannot define $\mathcal{O}_{L,S_L}$ in every finite extension L/k.*

Proof. Such a formula would have infinitely many inputs over k, since $\mathbb{Z} \subseteq \mathcal{O}_{k,S}$. Choose a rational prime p lying below no prime in S and such that $h(1/p) \neq 0$ for the polynomial in Theorem 2.1. There are infinitely many eligible primes and only finitely many roots of h. The theorem admits $1/p$ over some finite extension L/k. At every prime of L above p this element has negative valuation, and none of those primes belongs to S_L. Thus $1/p \notin \mathcal{O}_{L,S_L}$. □

These corollaries concern a fixed formula whose witness field is enlarged. They leave open the separate possibility of a different defining formula for each number field.

3 A sharper extension bound for one polynomial

Proposition 3.1. *Let k be an infinite field and $P \in k[T, Y_1, \ldots, Y_m]$. Suppose that P depends on the witness coordinate Y_j, and put $\delta = \deg_{Y_j} P \geq 1$. There is a nonzero $h \in k[T]$ such that for every $a \in k$ with $h(a) \neq 0$, the equation $P(a, \mathbf{y}) = 0$ has a solution over an extension of k of degree at most δ.*

Proof. Regard P as a polynomial in Y_j. Its leading coefficient is a nonzero polynomial in T and the remaining witness coordinates. Since k is infinite, choose constants for those remaining coordinates so that the leading coefficient becomes a nonzero polynomial $h(T)$. The specialization is a polynomial $Q(T, Z) \in k[T, Z]$ with $\deg_Z Q = \delta$. For $h(a) \neq 0$, choose any irreducible factor of the degree-δ polynomial $Q(a, Z)$. Adjoining a root gives an extension of degree at most δ and a witness for the original equation. □

Corollary 3.2. *Let k be a number field, and suppose a single nonzero polynomial $P(T, \mathbf{Y})$ has an infinite input projection over k. Then P depends on at least one witness coordinate. In Corollary 2.2, the bound on the extension degree can be taken to be*

$$D = \min_{j:\, \deg_{Y_j} P > 0} \deg_{Y_j} P \le \deg P.$$

In particular an unchanged quartic equation cannot define the rings of integers over every extension of degree at most four. If it has an essential witness of individual degree two, it already fails over an extension of degree at most two.

Proof. If P were independent of all witness coordinates, its input projection would be the finite zero set of a nonzero polynomial in T. Choose an essential coordinate of least positive individual degree, apply Proposition 3.1, and take any rational noninteger outside its finite exceptional set. The same argument applies to all such rational nonintegers. □

When the polynomial already defines an integral subset over k, the failure in the last sentence cannot occur over k itself; a witnessing extension of degree at most two must then have degree exactly two. This is an obstruction to transferring the equation unchanged, rather than a counterexample to field-dependent Diophantine definability.

An explicit quadratic extension

Let P_{392} be the quartic specified by the accompanying complete coefficient and residual tables. Coordinates are numbered $1, \dots, 392$, with input X_1. It is the sum of squares of 364 integer quadratic residuals. Put

$$A = 80989^{54929}, \qquad D = 16P_{392}(\tfrac{1}{2}, 0, \dots, 0).$$

This formula defines a fixed integer without requiring its decimal expansion. The exact computation in the supplied coefficient data gives

$$D = 191779334771151081400576\, A^2 + D_0, \qquad D_0 > 0,$$

where the complete decimal value of D_0 is recorded in `field-specialization392.json`.

Proposition 3.3. *Let $L = \mathbb{Q}(\theta)$ with $\theta^2 = -D$. Set $X_1 = 1/2$, $X_{293} = \theta/4$, and every other witness coordinate to zero. Then $P_{392} = 0$ over L, but X_1 is not an algebraic integer. Thus this unchanged quartic does not define $\mathcal{O}_L$ in L.*

Proof. Leave coordinate 293 free, calling it Y, and substitute the specified values in the quadratic residuals. Of the 364 residuals, 302 vanish, 61 become constants, and the residual with source identifier 493, the square-output definition, becomes $-Y$. Every constant belongs to $\frac{1}{4}\mathbb{Z}[A]$, and one has absolute value $1/4$. These statements follow by inspection of the explicitly listed integer terms; the separate checker also verifies them by exact substitution. Squaring and adding therefore gives

$$P_{392}(\tfrac{1}{2}\mathbf{e}_1 + Y\mathbf{e}_{293}) = Y^2 + D/16.$$

In particular D is a positive integer. The independent checker obtains this same identity both from all quadratic residuals and from the complete quartic coefficient table, including its fixed value of A. Since $D > 0$, the polynomial $Z^2 + D$ is irreducible over $\mathbb{Q}$, so L is an imaginary quadratic field. Taking $Y = \theta/4$ makes the displayed value zero. The rational number $1/2$ is not an algebraic integer, as proved in Corollary 2.2. □

References

[1] The Stacks Project Authors, *The Stacks Project*, Chevalley's theorem, Tag 00FE. https://stacks.math.columbia.edu/tag/00FE.